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HAWKEYE LAB

Blue Team Report

Network Forensics & Incident Documentation

Project

HAWKEYE

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Blue Team Report

Cover Information

Field	Detail
Project	HawkEye Lab Analysis
Author	Mohan Singh Parmar
Version	1.0
Date	2026-07-04
Classification	Internal / Portfolio Use
Source Evidence	<code>stealer.pcap</code> (4,003 packets, 2019-04-10 20:37:07 – 21:40:48 UTC)

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Executive Summary

On 2019-04-10, host **Beijing-5cd1-PC (10.4.10.132)**, a member of the **pizzajukebox.com** Active Directory domain, was infected with a Windows information-stealer after downloading an executable (**tkraw_Protected99.exe**) from an external HTTP server. Once resident, the malware performed recurring “what is my IP” beacon checks and exfiltrated data via automated SMTP sessions to an attacker-controlled mailbox roughly every ten minutes. This report documents the incident from a SOC/defensive perspective: how it could have been detected, the associated indicators of compromise, and the containment, eradication, and monitoring improvements recommended as a result.

Incident Overview

From a SOC perspective, this incident would most plausibly first surface as one or more of the following alerts, had the relevant detections been in place at the time:

- A proxy/firewall alert for an executable download from an uncategorized or newly observed domain (**proforma-invoices.com**).
- A DLP or egress-filtering alert for outbound SMTP (TCP/25) traffic originating from a non-mail-server endpoint.
- A threat-intelligence match on the destination IP **66.171.248.178** (**bot.whatismyipaddress.com**) or the SMTP server **23.229.162.69**, if either had prior reputation data.
- A UEBA/behavioral alert for a workstation making periodic, machine-generated HTTP requests at a fixed interval — a classic beacon signature.

None of the above alerting is confirmed to have fired in this specific case; this section describes the *detection opportunities* the incident should have generated.

Incident Identification

Attribute	Value
Affected host	Beijing-5cd1-PC (10.4.10.132)
Domain	pizzajukebox.com
Domain Controller	PizzaJukebox-DC (10.4.10.4)
Malicious file	tkraw_Protected99.exe
Delivery source	proforma-invoices.com (217.182.138.150)
Exfiltration destination	SMTP server 23.229.162.69:25
Exfil account/domain	sales.del@macwinlogistics.in
Beacon destination	bot.whatismyipaddress.com (66.171.248.178)
First observed malicious activity	2019-04-10 20:37:54 UTC (malware download)
Classification	Information stealer / credential exfiltration

Detection Process

SIEM

Replace this section with your actual HawkEye Lab SIEM platform and specific detection rule names/IDs used to alert on this activity.

IDS/IPS

A network IDS with an up-to-date ruleset (e.g., Suricata/Snort with ET Open or a commercial feed) should have flagged:

- The HTTP download of a Windows PE executable from a domain with no prior reputation, particularly given the mismatched/outdated **MSIE 7.0** user-agent string.
- SMTP **AUTH LOGIN** traffic originating from a workstation IP rather than an approved mail relay/gateway.

Firewall

Egress filtering logs would show:

- Repeated outbound TCP/80 connections from 10.4.10.132 to 66.171.248.178 at ~10 minute intervals.
- Repeated outbound TCP/25 connections from 10.4.10.132 to 23.229.162.69, which should be unusual if the firewall policy restricts direct SMTP egress to an approved mail relay only.

Endpoint / Antivirus

Host-based AV/EDR should be checked for:

- Detection or quarantine events for the SHA256 62099532750dad1054b127689680c38590033fa0bdfa4fb40c7b4dcb2607fb11 / MD5 71826ba081e303866ce2a2534491a2f7 .
- Process creation and network-connection telemetry correlating tkraw_Protected99.exe (or its in-memory/renamed process) to the outbound HTTP and SMTP connections identified in this pcap.

Replace this subsection with actual AV/EDR console findings if available from your HawkEye Lab environment; this pcap alone cannot confirm host-side detection status.

Windows Logs

Relevant Windows Event Log sources to review on Beijing-5cd1-PC:

- Security Event ID 4688 (process creation) around 20:37:54 UTC for execution of the downloaded file.
- Sysmon Event ID 1 (process creation), Event ID 3 (network connection) for correlating the beacon and SMTP sessions to a specific process.
- Sysmon Event ID 11/23 (file creation) for the dropped executable.

Linux Logs

Not applicable to the victim host in this capture (Windows workstation); however, if the SMTP relay or web proxy logs are hosted on Linux infrastructure, review /var/log/squid (or equivalent) and mail relay logs for corroborating entries.

Indicators of Compromise

IP Addresses

IP	Role
217.182.138.150	Malware delivery host (proforma-invoices.com)

IP	Role
66.171.248.178	Beacon/recon destination (bot.whatismyipaddress.com)
23.229.162.69	Exfiltration SMTP server
10.4.10.132	Infected internal host (Beijing-5cd1-PC)
10.4.10.4	Internal Domain Controller (PizzaJukebox-D0 — legitimate, not an IOC

Domains

Domain	Role
proforma-invoices.com	Malware hosting/delivery
macwinlogistics.in	Exfiltration mailbox domain
bot.whatismyipaddress.com	Third-party IP-check service abused for recon

URLs

URL	Role
http://proforma-invoices.com/proforma/tkraw_Protected99.exe	Malicious payload delivery URL

File Names / Hashes

Attribute	Value
File name	tkraw_Protected99.exe
File size	2,025,472 bytes
MD5	71826ba081e303866ce2a2534491a2f7
SHA256	62099532750dad1054b127689680c38590033fa0bdfa4fb40c7b4dcb2607fb11

Processes

Replace this section with the actual process name/PID confirmed via endpoint telemetry (Sysmon/EDR) to have made the outbound connections; not derivable from network capture alone.

Registry Keys

Replace this section with any persistence-related registry keys (e.g., Run/RunOnce keys) identified via host forensics; no registry activity is visible in a network capture.

User Accounts

Account	Context
beijing-5cd1-pc\$	Domain computer account used for Kerberos authentication (legitimate)
sales.del@macwinlogistics.in	Attacker-controlled SMTP account used for exfiltration AUTH LOGIN and as the mail envelope sender/recipient

Log Analysis

Authentication Logs

The only authentication activity visible is legitimate Kerberos AS-REQ/AS-REP and TGS-REQ/TGS-REP exchanges between 10.4.10.132 and the Domain Controller, consistent with normal domain logon/service-ticket behavior. No anomalous authentication (e.g., unusual logon type, off-hours interactive logon) is evidenced in this capture.

Event Logs

Beijing-5cd1-pc\$ stamps in this report align with local Windows Security and Sysmon event logs on Beijing-5cd1-pc\$ stamps in this report align with local Windows Security and Sysmon event logs on

Firewall Logs

Cross-reference perimeter firewall/proxy logs for the destination IPs/domains listed under Indicators of Compromise to confirm allow/deny decisions at the time of the incident and to search for other internal hosts contacting the same infrastructure.

Syslogs

Replace this section with relevant syslog entries from network infrastructure (DNS resolvers, mail relays) if centralized logging was in place for this HawkEye Lab environment.

Web Logs

If a web/HTTP proxy was in place, its logs should show the `GET /proforma/tkraw_Protected99.exe` request and the recurring `GET /` requests to `bot.whatismyipaddress.com`; these are currently only confirmed via the pcap itself.

Incident Timeline

Time (UTC)	Event
20:37:54	Malicious executable <code>tkraw_Protected99.exe</code> downloaded via HTTP from <code>proforma-invoices.com</code>
20:38:15	First beacon/recon <code>GET /</code> request to <code>bot.whatismyipaddress.com</code>
20:38:16	First SMTP exfiltration session established to <code>23.229.162.69:25</code>
20:48:20	Second beacon and SMTP exfiltration cycle
20:58:24	Third beacon and SMTP exfiltration cycle
21:08:30	Fourth beacon (<code>bot.whatismyipaddress.com</code>)
21:18:34	Fifth beacon (<code>bot.whatismyipaddress.com</code>)
21:28:38	Beacon and SMTP exfiltration cycle
21:38:42	Final beacon and SMTP exfiltration cycle observed in capture
21:40:48	Capture ends; incident status at this point unresolved/ongoing

MITRE ATT&CK Mapping

Tactic	Technique	Description
Initial Access	T1189 – Drive-by Compromise / T1566 – Phishing (inferred)	Executable delivered via HTTP from a lure-themed domain; the initiating email/link itself is not

Tactic	Technique	Description
		present in this capture but is the most likely delivery vector
Execution	T1204.002 – User Execution: Malicious File	Downloaded .exe presumed executed by the user or a script, based on subsequent malicious network activity
Discovery	T1016 – System Network Configuration Discovery	Recurring queries to a public “what is my IP” service to fingerprint the victim’s external address
Command and Control	T1071.001 – Application Layer Protocol: Web Protocols	Use of standard HTTP for beacon/ check-in traffic
Exfiltration	T1048.003 – Exfiltration Over Unencrypted/Obfuscated Non-C2 Protocol	Use of SMTP to exfiltrate data to an attacker-controlled mailbox
Collection (inferred)	T1056 – Input Capture / T1113 – Screen Capture	Not directly observable in network traffic, but consistent with the known behavior of information-stealer/keylogger malware families using this delivery-and-SMTP-exfil pattern

Detection Opportunities

1. **New/rare domain executable download** — alert on any .exe/.dll/script download from a domain with no established reputation or first-seen within the last N days.
2. **Fixed-interval beaconing** — alert on any internal host making outbound HTTP requests to the same external host at a highly regular interval (here, ~10 minutes) over a sustained period.
3. **Direct SMTP egress from non-mail-infrastructure hosts** — alert (or block) any workstation-originated TCP/25 connection that does not traverse the approved internal mail relay.
4. **User-Agent anomaly** — alert on HTTP requests using outdated/inconsistent user-agent strings (e.g., MSIE 7.0 from a host whose other traffic indicates a modern OS/browser).
5. **DNS-based detection** — alert on DNS queries for macwinlogistics.in or other domains matching known infostealer exfiltration infrastructure once added to threat-intel feeds.

Containment

1. Immediately isolate **Beijing-5cd1-PC** from the network (NAC quarantine VLAN or switch port disable) to stop further exfiltration.
 2. Block outbound traffic to **217.182.138.150**, **66.171.248.178**, and **23.229.162.69** at the perimeter firewall.
 3. Block DNS resolution for **proforma-invoices.com**, **macwinlogistics.in**, and any related infrastructure via DNS sinkhole/RPZ.
 4. Preserve full volatile memory and disk image of the affected host before remediation for forensic completeness.
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Eradication

1. Remove the malicious executable and any dropped/persistence artifacts from **Beijing-5cd1-PC** (reimage is strongly preferred over manual cleanup for commodity stealer malware).
 2. Rotate credentials for any accounts that may have been captured (local Windows account, browser-stored credentials, and any credentials typed on the host during the infection window).
 3. Revoke and reissue Kerberos tickets / force password reset for the computer account if domain-integrity concerns are identified during host forensics.
 4. Confirm removal via a full AV/EDR scan and hash-based verification against **62099532750dad1054b127689680c38590033fa0bdfa4fb40c7b4dcb2607fb11**.
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Recovery

1. Reimage and rejoin **Beijing-5cd1-PC** to the domain from a known-good, patched baseline image.
 2. Monitor the host closely (enhanced logging/EDR sensitivity) for 30 days post-recovery for any reinfection indicators.
 3. Notify affected users to change passwords for any personal or business accounts accessed from the infected host during the compromise window.
 4. Conduct a review of any business email/financial transactions the user may have initiated during the infection period, given the “invoice”-themed lure filename suggests possible Business Email Compromise (BEC) targeting.
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Monitoring Recommendations

- Deploy egress filtering rules restricting SMTP (TCP/25) from all endpoints except the approved mail relay.
 - Add the IOCs identified in this report to SIEM watchlists and threat-intelligence platform (TIP) feeds.
 - Implement beacon-detection analytics (regular-interval outbound connection detection) in the SIEM/NDR platform.
 - Enable full HTTP object logging/extraction on the web proxy for retrospective analysis of future incidents.
 - Enable Sysmon (or equivalent EDR telemetry) fleet-wide, at minimum for process creation (Event ID 1) and network connection (Event ID 3) logging.
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Security Improvements

1. Implement application allow-listing to prevent unapproved executables from running on end-user workstations.
 2. Deploy web/email gateway sandboxing to detonate suspicious attachments/downloads before delivery to the user.
 3. Enforce SMTP egress control at the firewall (block direct outbound TCP/25 from all non-relay hosts).
 4. Expand user security-awareness training to cover invoice-themed phishing lures specifically.
 5. Integrate threat-intelligence feeds for newly registered/low-reputation domains into the web proxy and DNS filtering layer.
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Lessons Learned

- The absence of SMTP egress restrictions allowed a single infected workstation to directly exfiltrate data to an external mail server, bypassing any DLP or content inspection that would normally sit on the corporate mail gateway.
 - Regular-interval beacon traffic, even over benign-looking protocols (plain HTTP **GET /**), is a highly reliable behavioral indicator that should be a standard detection rule regardless of payload content.
 - Full packet capture proved sufficient to reconstruct the entire incident timeline in the absence of endpoint telemetry, reinforcing the value of network-level visibility as a compensating control.
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Conclusion

This incident reflects a common but serious information-stealer compromise: a single user-driven executable download led to sustained, automated data exfiltration via SMTP over more than an hour of observed traffic. The absence of egress controls on SMTP and executable downloads were the primary contributing gaps. Implementing the containment, eradication, and monitoring recommendations in this report — particularly SMTP egress restriction and beacon-detection analytics — would have significantly reduced both the likelihood and impact of this incident.

References

- MITRE ATT&CK Framework: <https://attack.mitre.org/>
- Wireshark/**tshark** documentation: <https://www.wireshark.org/docs/>
- Replace this section with internal SOC runbooks, threat-intel platform links, or additional HawkEye Lab reference material used during this investigation.